DEMON: Dynamic Exploration of Molecular Dynamics Simulations Through Optimized Navigation

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Introduction

Molecular dynamics (MD) models atomic motion to study processes such as protein conformational changes and drug binding. However, high free-energy barriers place many biologically important transitions beyond conventional simulation timescales.[1] Enhanced-sampling methods accelerate exploration but often rely on predefined reaction coordinates or system-specific features.[2] We propose DEMON, a reinforcement learning framework to decide which molecular trajectories in an ensemble are the most important for future exploration, specifically targeting cross-system transferability. DEMON replicates an agent-selected trajectory and randomly prunes others to maintain a fixed ensemble size. It learns time series representations directly from raw molecular coordinates.

Why the MD Game Environment

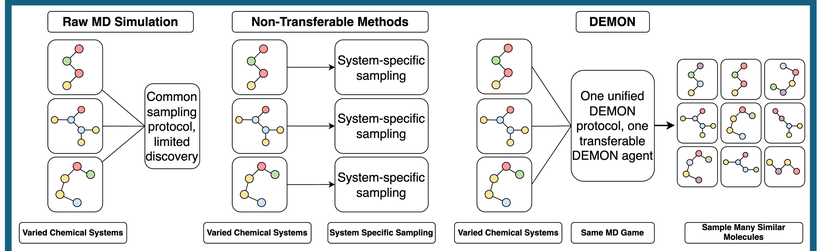


Figure 1: Most biomolecular simulation is intractable for raw MD simulation. Many enhanced sampling methods either require system specific knowledge or fit to a specific system, limiting their efficacy on novel systems. DEMON learns an exploration policy that transfers between systems like those seen in training.

The Ensemble Split-Prune Environment

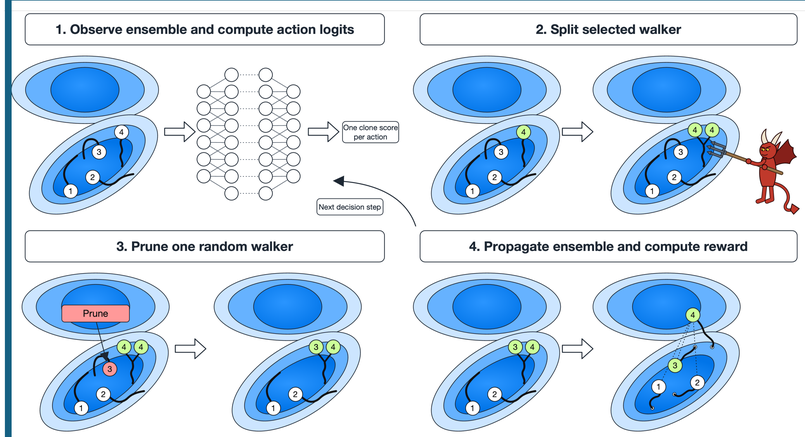


Figure 2: DEMON formulates ensemble sampling as a reinforcement learning problem. The agent repeatedly reallocates a fixed simulation budget toward selected trajectories, using the resulting diversity reward to improve future decisions.

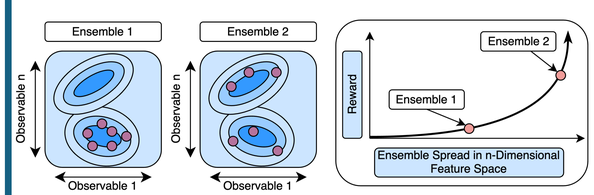


Figure 3: A current reward function candidate quantifies the diversity of the walker using their cartesian separation in an n-dimensional feature space. It is not necessary that the observables are strong CVs, but it is necessary that there is some correlation between the combination of observables and the different states of the system.

Molecular Time Series Representations

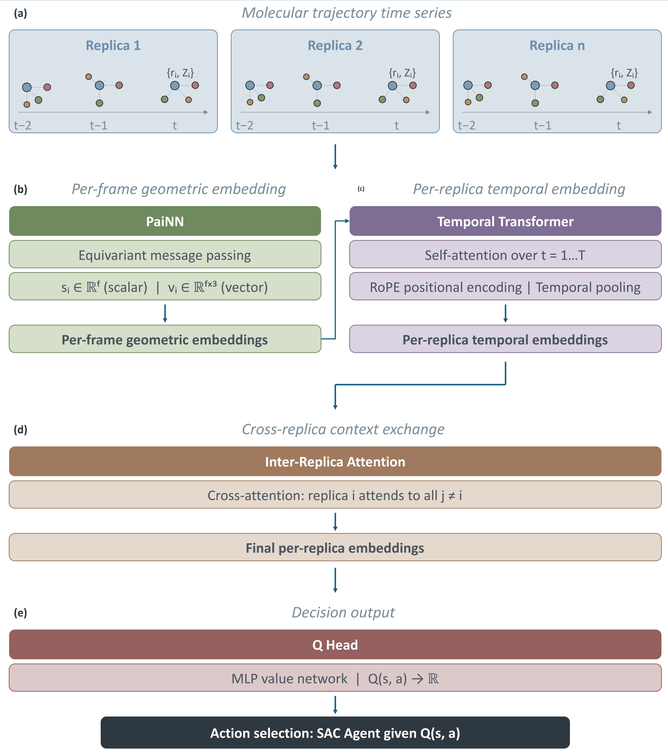


Figure 4: (a) DEMON processes raw molecular trajectories using PaiNN[4], **(b)** attends over trajectory frames with a temporal transformer[5], **(d)** attends between replicas for ensemble context, and **(e)** maps final representations to action probabilities.

Model System Results

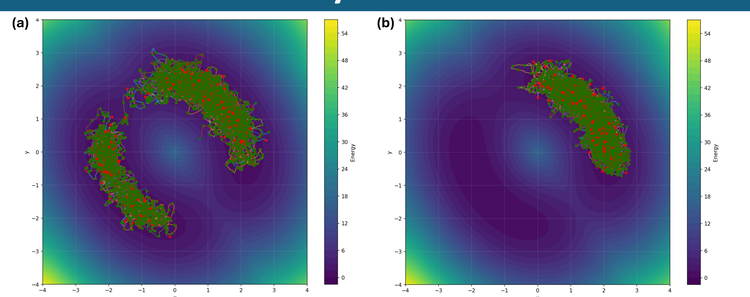


Figure 5: DEMON transfers to out of distribution model systems. Evaluation trajectories from **(a)** a DEMON policy and **(b)** random splitting using equal compute budgets on an out-of-distribution environment. DEMON explores both basins while random splitting remains in the initial basin.

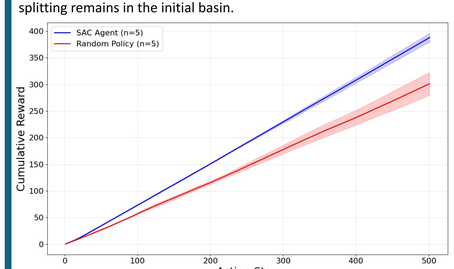


Figure 6: DEMON achieves higher cumulative exploration reward (mean $\pm$1SD; $n = 5$) than random splitting on a random OOD 8-dimensional multi-basin GMM landscape under equal compute and environment parameters

Biomolecular Results

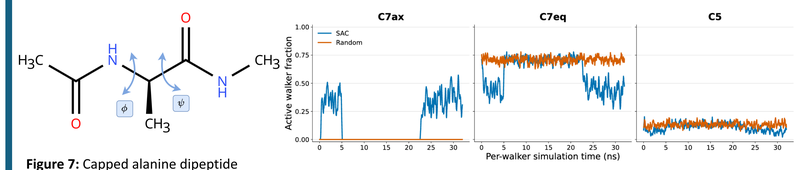


Figure 7: Capped alanine dipeptide structure (Ace-Ala-Nme; Ala2) with labeled ϕ and ψ dihedral angles. The molecule is simulated in a vacuum using implicit solvation simulation.

Figure 8: Fraction of current DEMON (blue) and random (orange) walkers in each basin over per-walker simulation time. DEMON achieves transitions to the rare C7ax basin and C5 basin while random splitting only samples the initial C7eq basin and C5 basin.

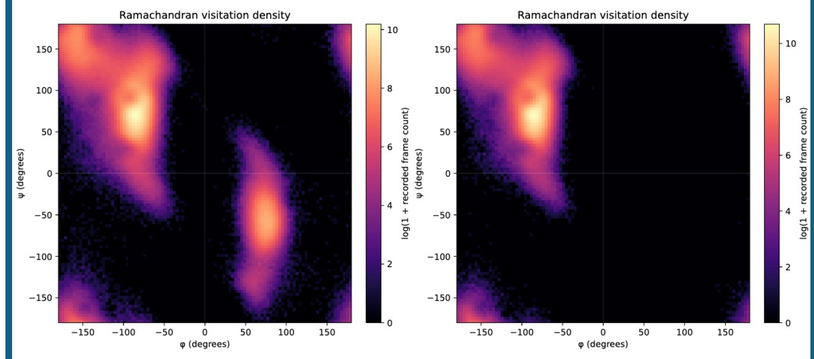


Figure 9: DEMON **(a)** samples a broader set of conformations than random splitting **(b)** in exploration of Ala2 in a vacuum at 350 K. DEMON explores the C_{7eq}, C_{7ax}, and C_5 conformations, while random splitting only explores C_{7eq} and C_5 conformations with equal computational budget.

Conclusions & Future Work

DEMON improves exploration across model systems. DEMON consistently outperforms random splitting and transfers zero-shot to unseen landscape configurations.

Early atomistic results are positive. DEMON achieves greater configurational diversity than random splitting on Ala2. Performance at the target operating temperature of 300 K remains a challenge.

Address rare events during training. Rare transitions occur too infrequently for the agent to learn effective allocation at low temperatures. How could curriculum learning or replay biasing impact this?

Aim for transferability on unseen peptide systems. The next core goal of DEMON is to learn a policy that transfers across small OOD polypeptide systems.

Reproduce accurate kinetics. Between decision points, each walker runs unbiased MD, so standard estimators or reweighting can be applied to the resulting trajectory segments. The next goal is to demonstrate that DEMON's short trajectories yield accurate thermodynamic and kinetic quantities.

References

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